Exercise 1

A paper published by *Quality Engineering* reported a dataset that consists of loading weigths (in grams) of insecticide tanks. See *insecticide_tanks.csv*

x1	*x2*	*x3*	*x4*
456	458	439	448
459	462	495	500
443	453	457	458
470	450	478	470
457	456	460	457
434	424	428	438
460	444	450	463
467	476	485	474
471	469	487	476
473	452	449	449
477	511	495	508
458	437	452	447
427	443	457	485
491	463	466	459
471	472	472	481
443	460	462	479
461	476	478	454

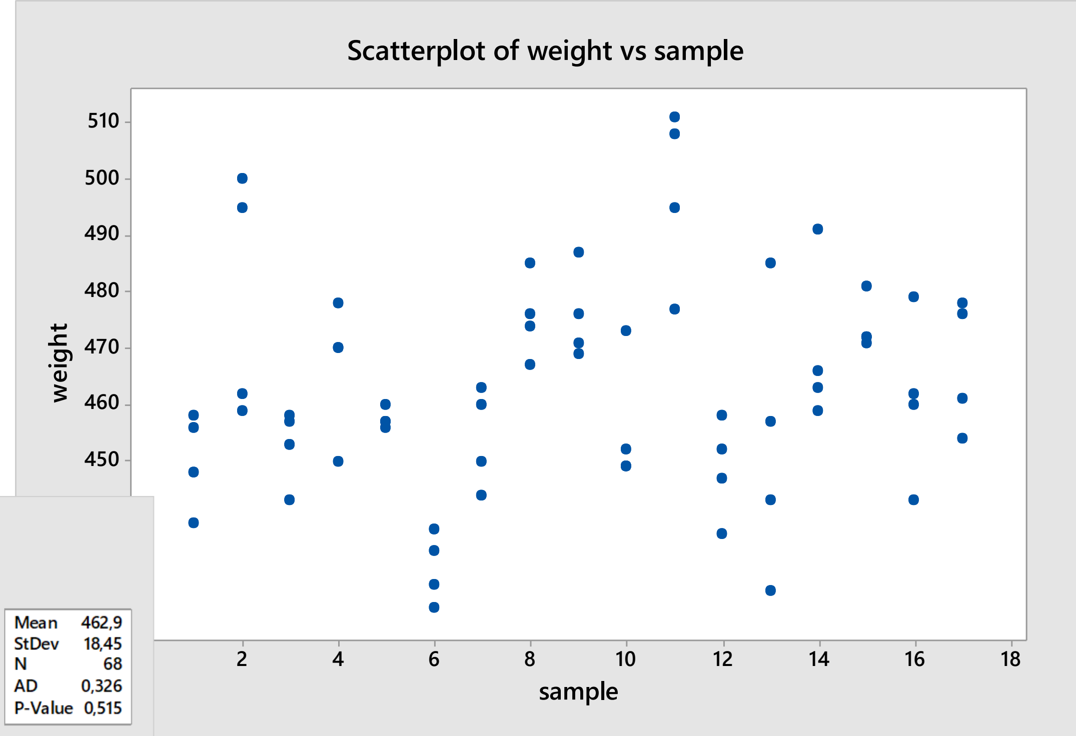
a) Design a control chart for the mean. Which conclusions can we draw about the process?

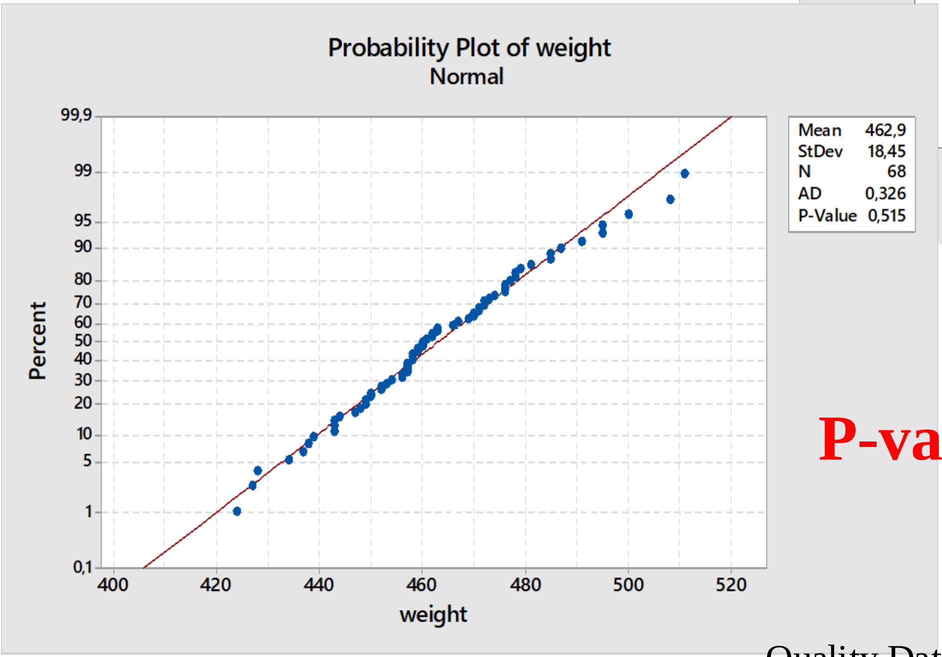
b) Consider the sample means as invidual measurements: design a control chart for the mean

c) Compare the results at point a) and b). What can we conclude?

Exercise 1 (solution)

a) Graphical analysis of data





P-value=0.515

Exercise 1 (solution)

xmean	R
450.25	19
479.00	41
452.75	15
467.00	28
457.50	4
431.00	14
454.25	19
475.50	18
475.75	18
455.75	24
497.75	34
448.50	21
453.00	58
469.75	32
474.00	10
461.00	36
467.25	24

a) Let's design a Xbar-R control chart

Descriptive Statistics: weight; range

Variable	Mean	SE Mean	StDev	Q1	Median	Q3
weight	462,94	2,24	18,45	450,50	460,00	475,50
range	24,41	3,14	12,93	16,50	21,00	33,00

$\overline{X}$ Control Chart

$$\begin{cases} UCL = \overline{\overline{X}} + A_2(n)\overline{R} \\ CL = \overline{\overline{X}} \\ LCL = \overline{\overline{X}} - A_2(n)\overline{R} \end{cases}$$

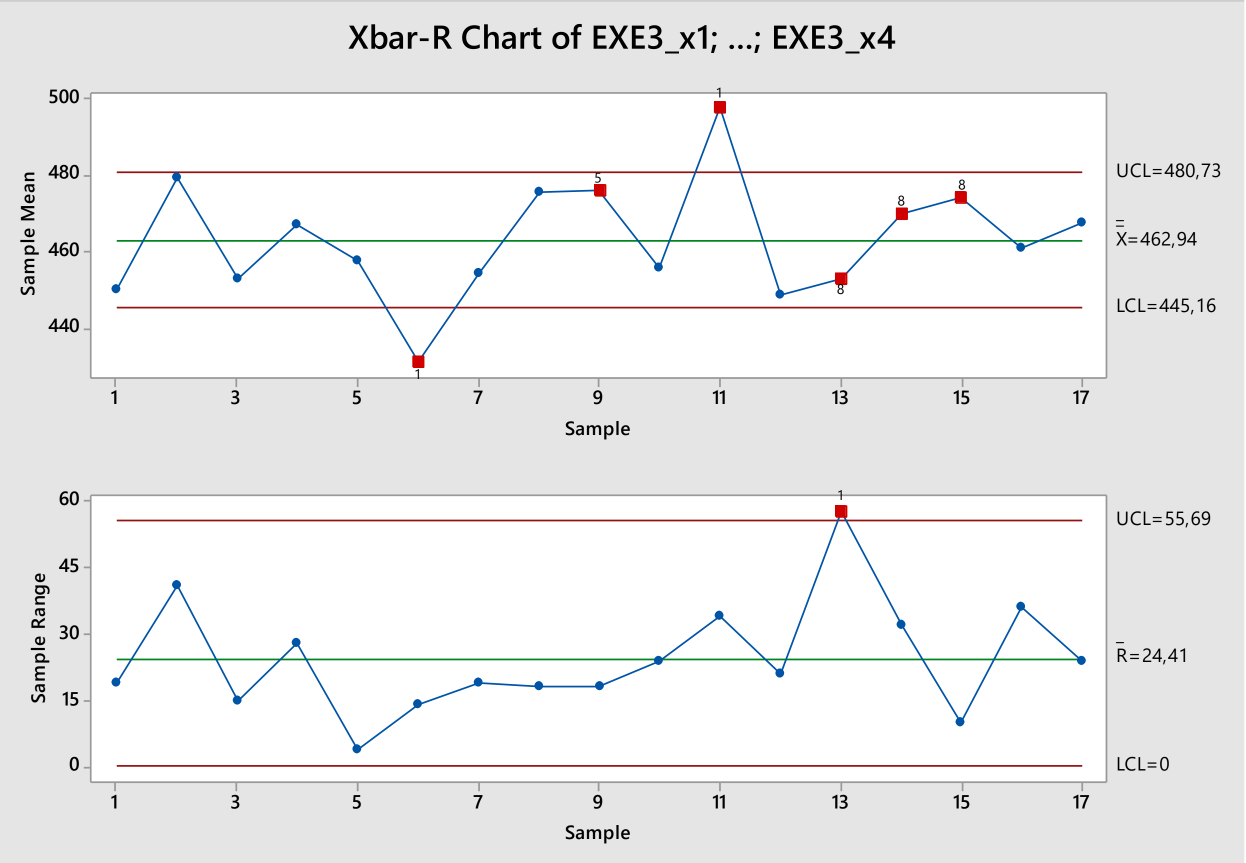
R Control Chart

$$\begin{cases} UCL = D_4(n)\overline{R} \\ CL = \overline{R} \\ LCL = D_3(n)\overline{R} \end{cases}$$

$$n = 5$$

Exercise 1 (solution)

a)



Test Results for Xbar Chart of EXE3_x1; ...; EXE3_x4

```
TEST 1. One point more than 3,00 standard deviations from center line.
Test Failed at points:  6; 11
TEST 5. 2 out of 3 points more than 2 standard deviations from center line (on one side of CL).
Test Failed at points:  9; 11
TEST 8. 8 points in a row more than 1 standard deviation from center line (above and below CL).
Test Failed at points:  13; 14; 15
```

Test Results for R Chart of EXE3_x1; ...; EXE3_x4

```
TEST 1. One point more than 3,00 standard deviations from center line.
Test Failed at points:  13
```

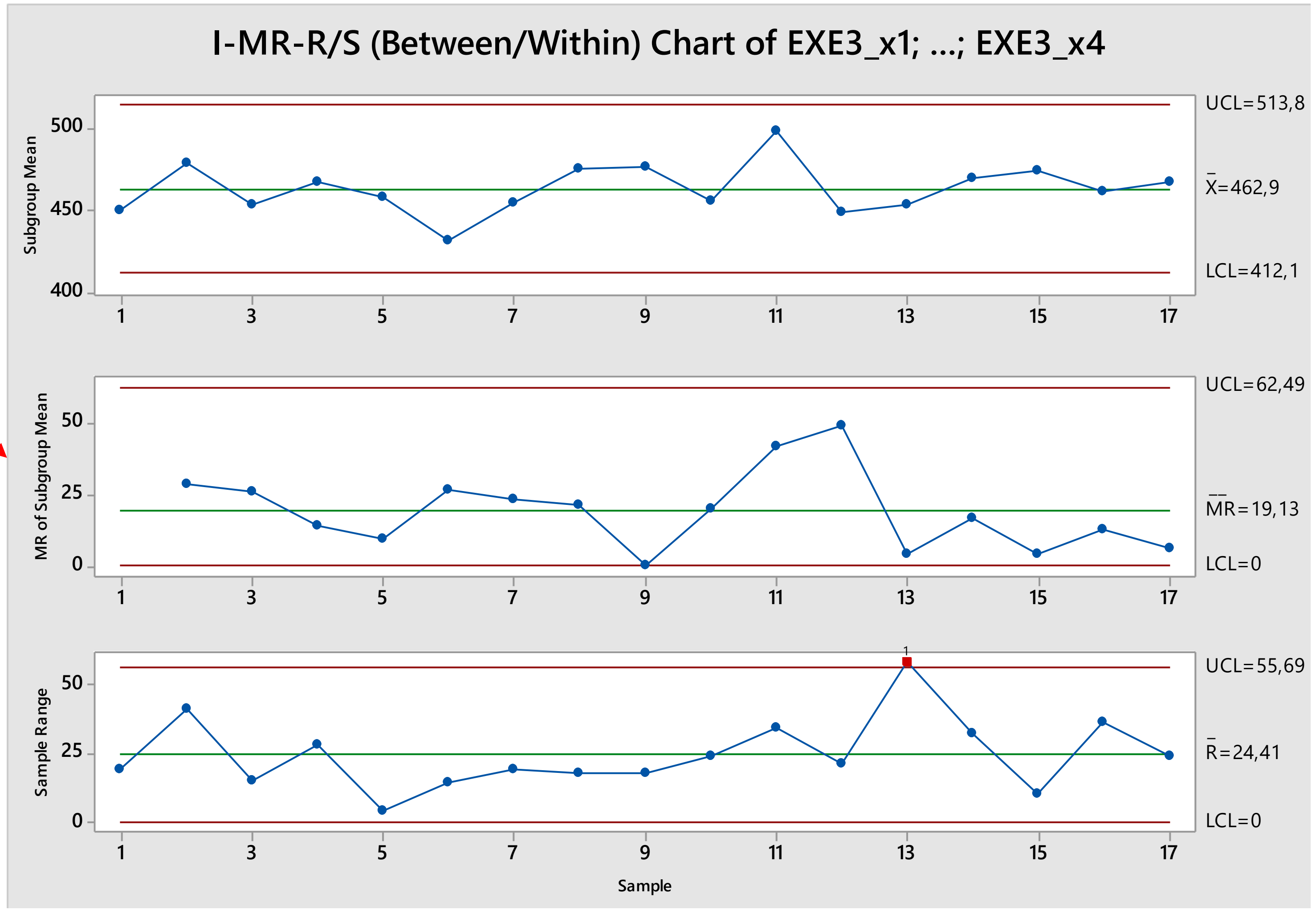
b) Exercise 1 (solution)

Design a “Between groups” control chart, i.e., a chart that assumes all the samples to be individual measures

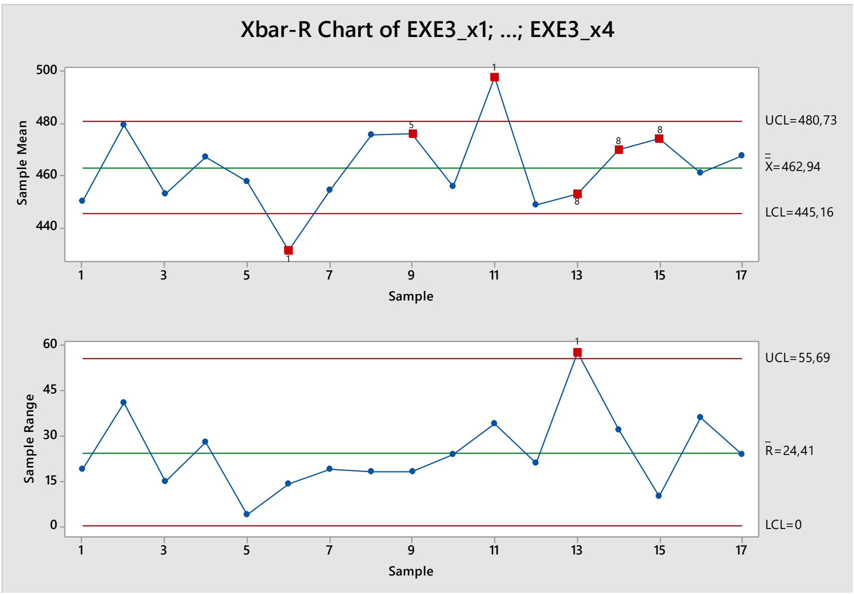
MINITAB Command:

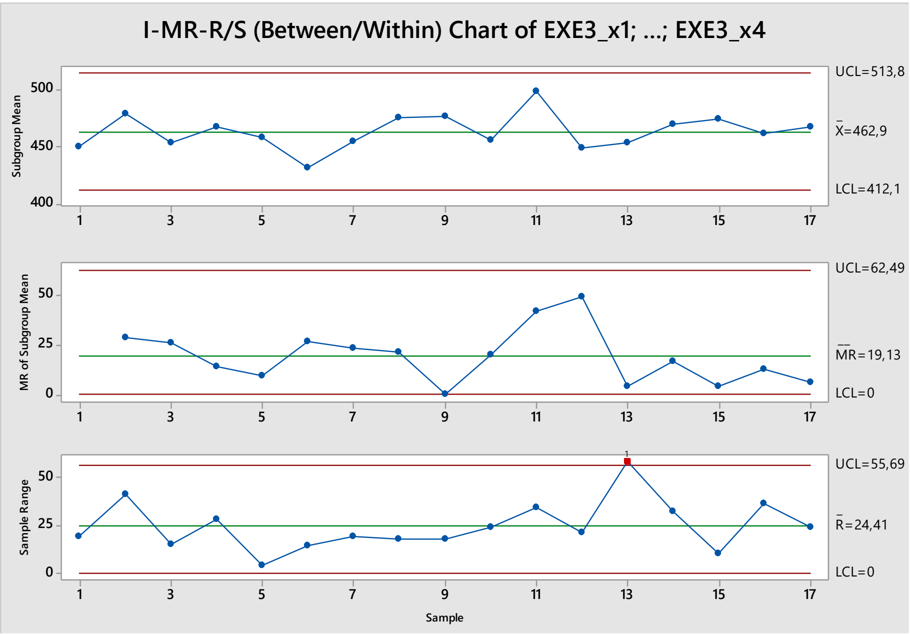
Stat->Control charts->Variable Charts for Subgroups->I-MR-R/S (Between/Within).

MR of subgroup means



c)





From the results of point 1 we can conclude that the Xbar-R chart is not appropriate to monitor these data. The I-MR-R chart designed in point 2 is more effective, as it allows to monitor the process mean and the variability of the process. The I-MR-R chart is more robust with respect to the violation of the independence assumption within the sample.

EXERCISE 2

The table reports life-time measurements for eletric circuits.
See *electric_circuits.csv*.

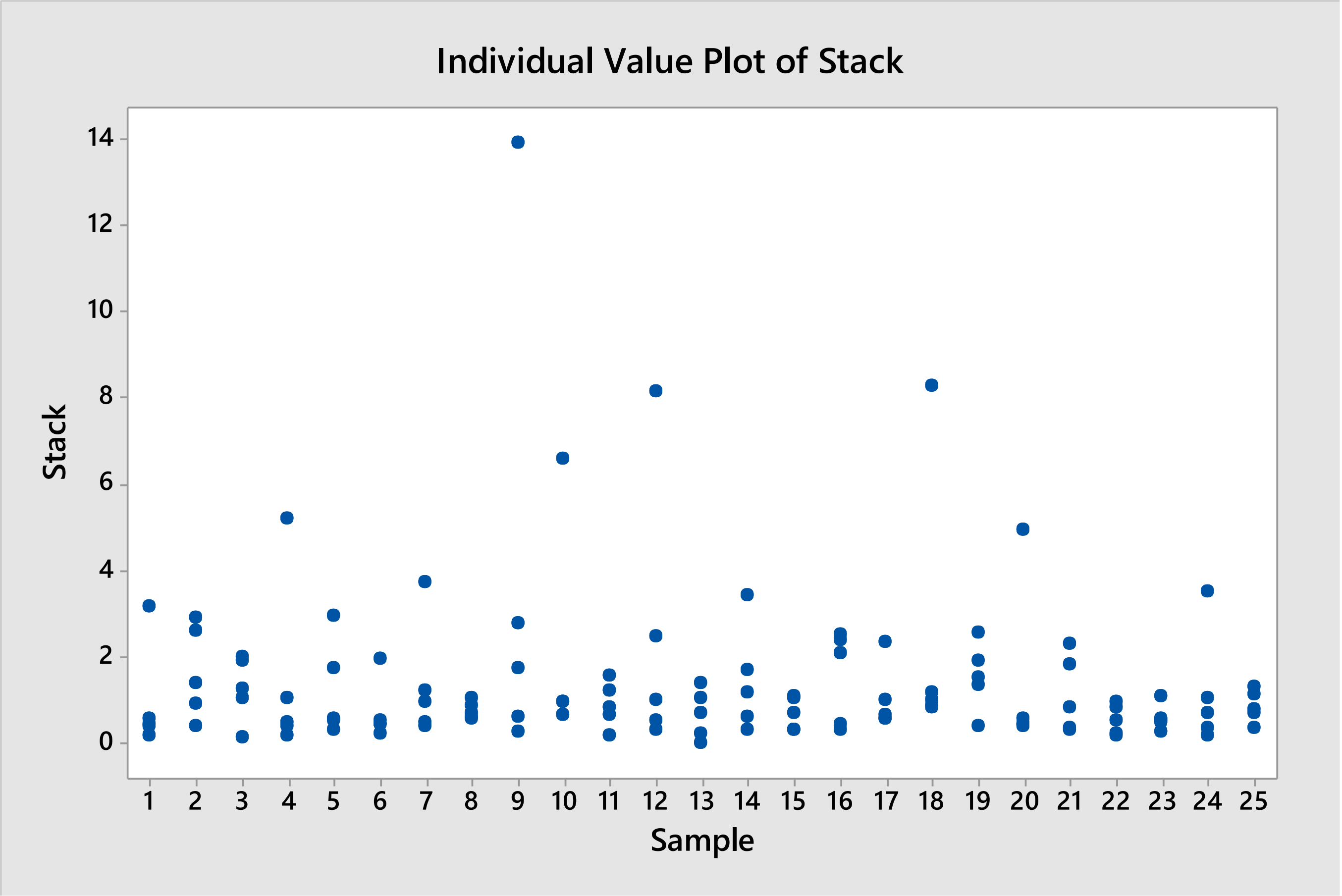
Design a $\bar{X} - R$ control chart.
Which conclusions can be drawn about the process?

x1	x2	x3	x4	x5
0.473	0.405	0.213	3.187	0.572
0.430	2.623	1.415	0.915	2.933
0.148	1.938	1.057	2.019	1.256
5.209	0.211	1.047	0.492	0.388
0.308	0.536	0.570	2.951	1.741
0.539	0.229	1.969	0.449	0.461
3.763	0.500	0.954	0.407	1.250
0.705	0.571	1.038	0.880	0.607
13.915	0.300	2.812	1.735	0.644
0.666	0.961	0.659	6.590	0.984
1.223	1.590	0.828	0.200	0.668
0.532	0.343	2.470	8.159	0.996
0.721	0.029	1.392	1.073	0.235
0.605	1.185	1.706	0.339	3.455
1.114	0.323	1.054	0.303	0.716
2.535	2.381	2.087	0.463	0.301
2.365	0.576	0.680	1.028	0.603
0.831	0.994	1.175	0.867	8.310
1.937	1.553	0.415	2.581	1.361
4.969	0.449	0.424	0.457	0.570
0.840	0.323	1.843	2.316	0.360
0.852	0.176	0.556	0.988	0.248
0.271	0.517	0.592	1.104	0.544
1.080	0.344	0.177	0.716	3.526
0.805	1.162	1.306	0.733	0.359

EXERCISE 2 (SOLUTION)

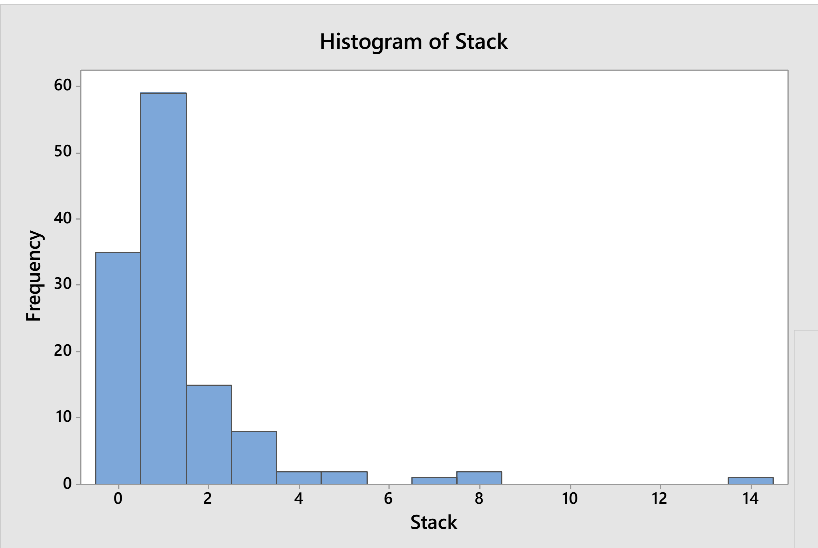
DATA SNOOPING

Outliers are present

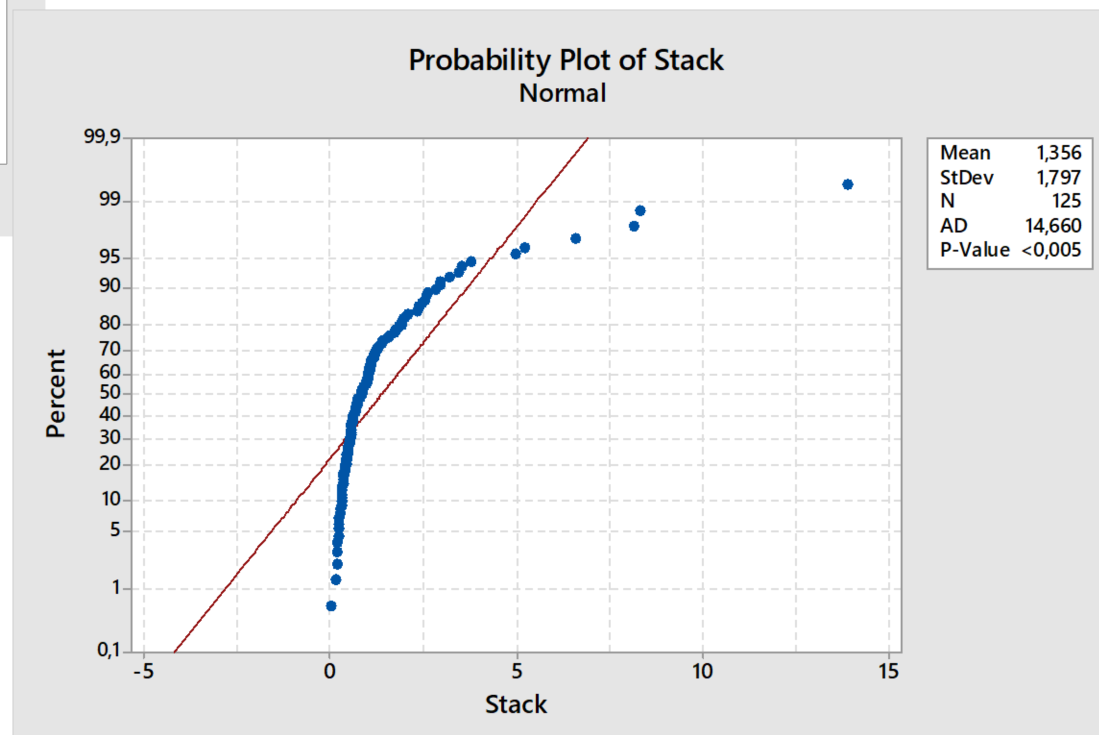


EXERCISE 2 (SOLUTION)

DATA SNOOPING



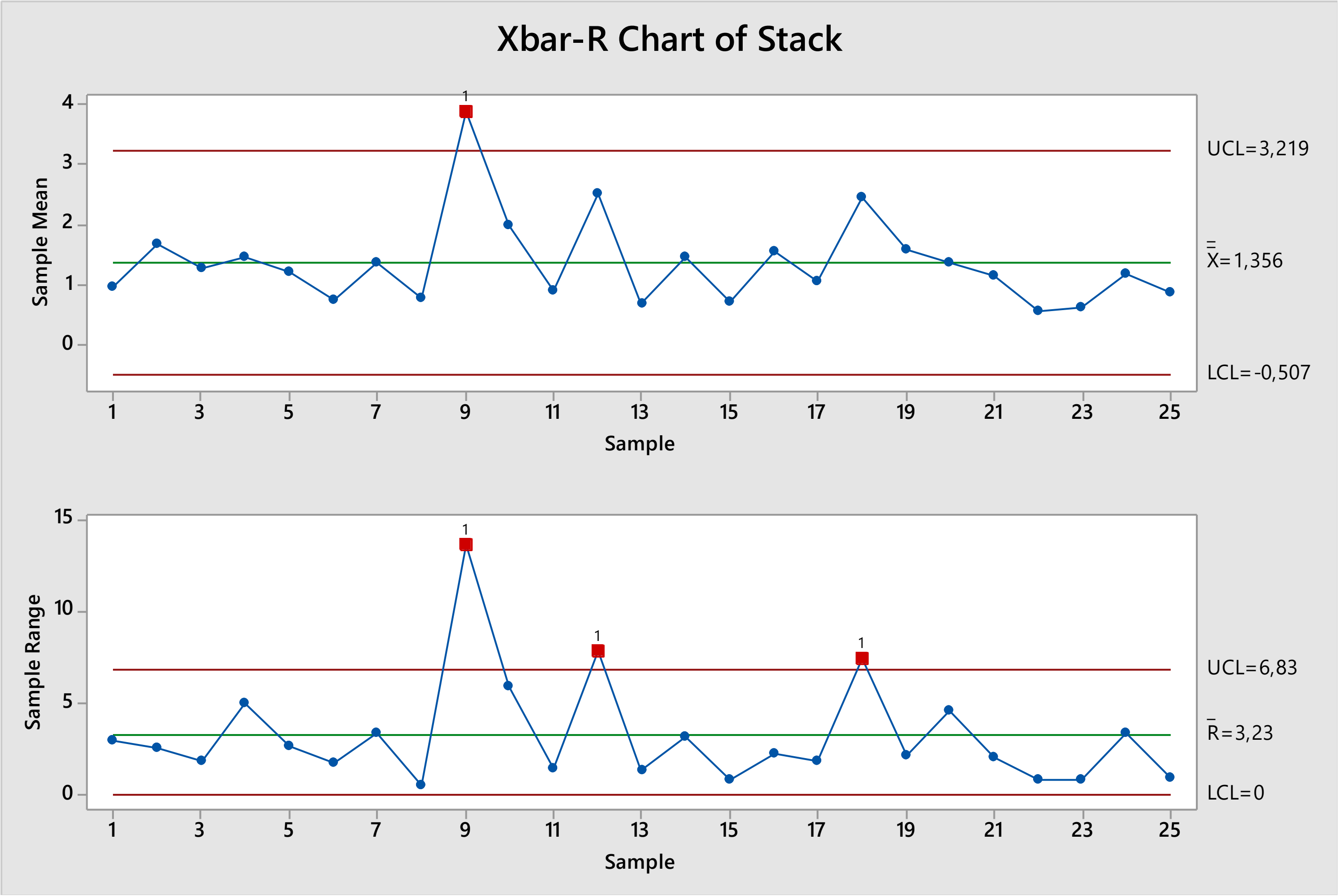
Skewed distribution



Non-normal data

EXERCISE 2 (SOLUTION)

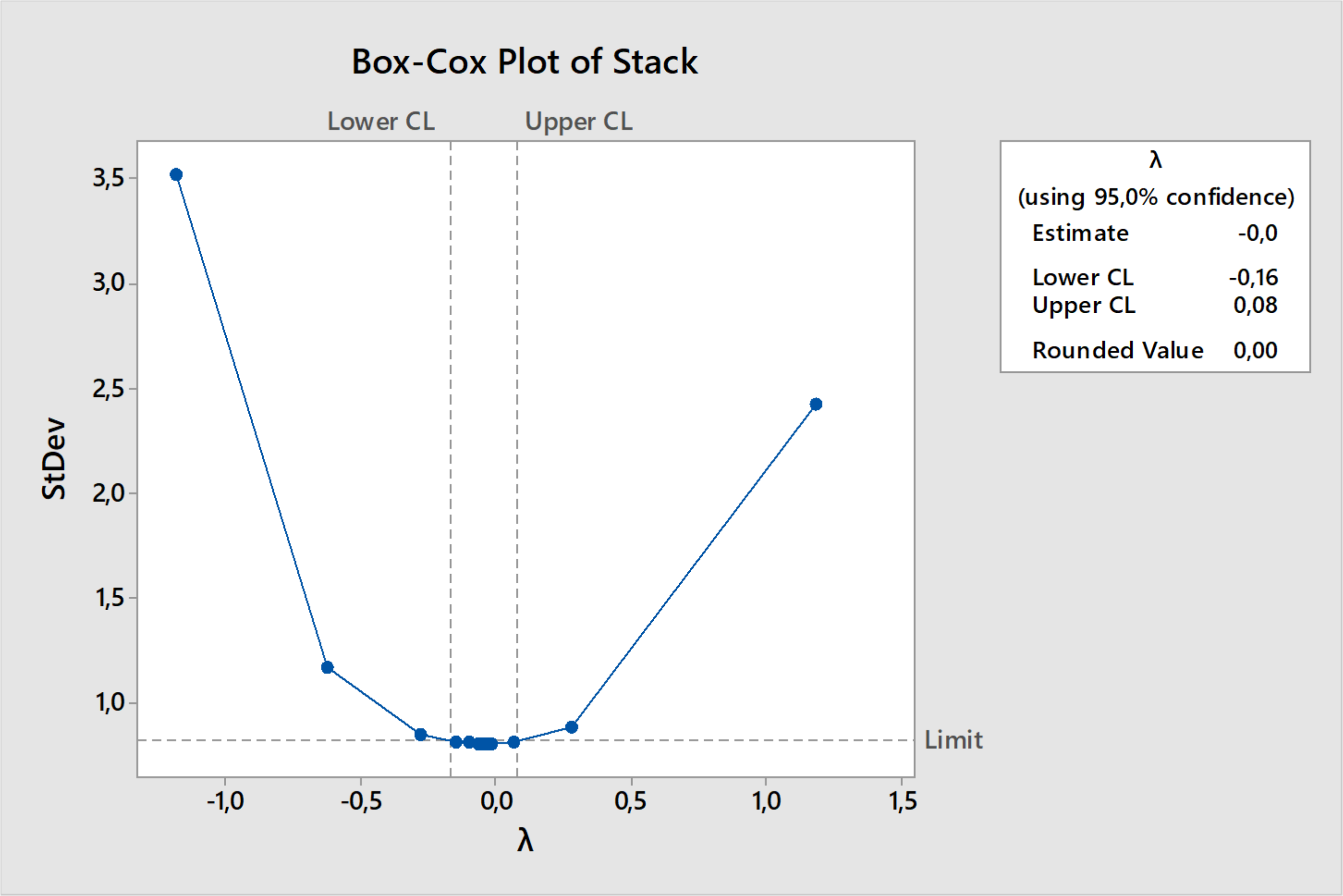
What happens if we neglect the normality violation?



OOC observations may be due to a violation of control chart assumptions!

EXERCISE 2 (SOLUTION)

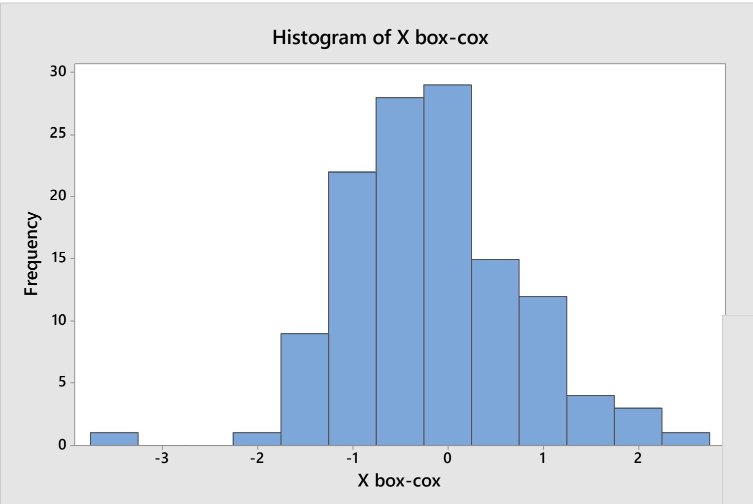
Let's try to transform the data…



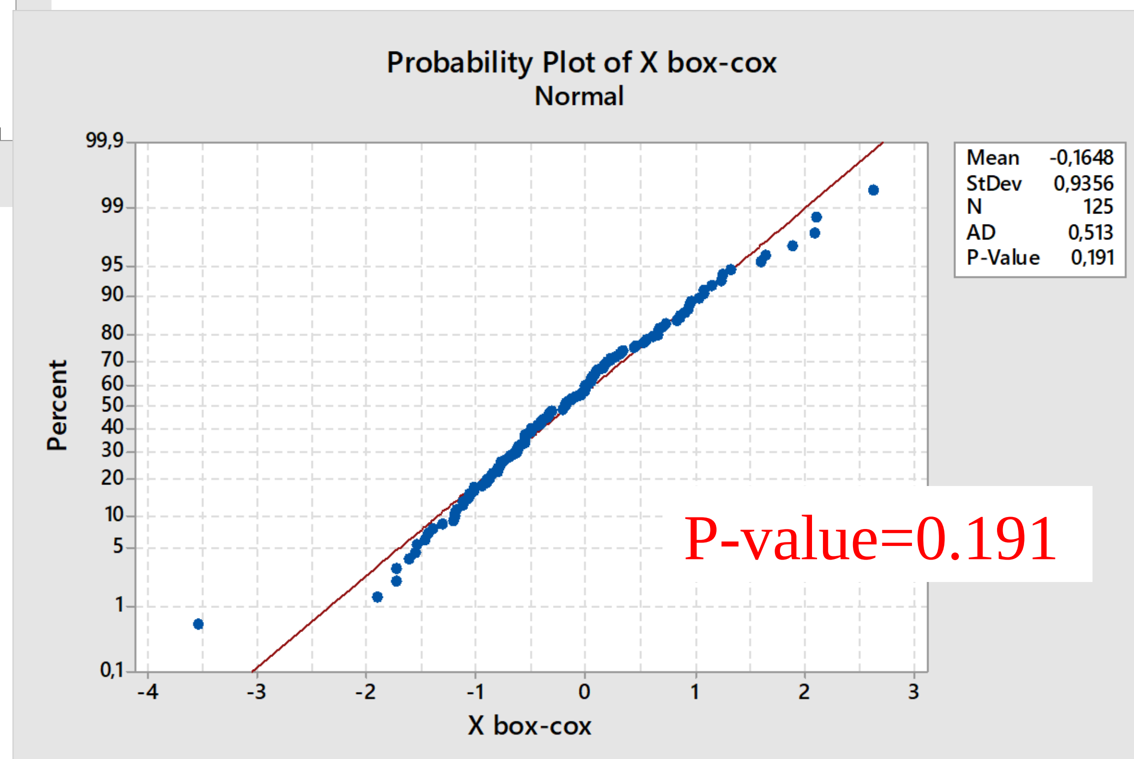
Choose $\lambda = 0$, i.e. the logarithm transformation

EXERCISE 2 (SOLUTION)

Check normality for transformed data



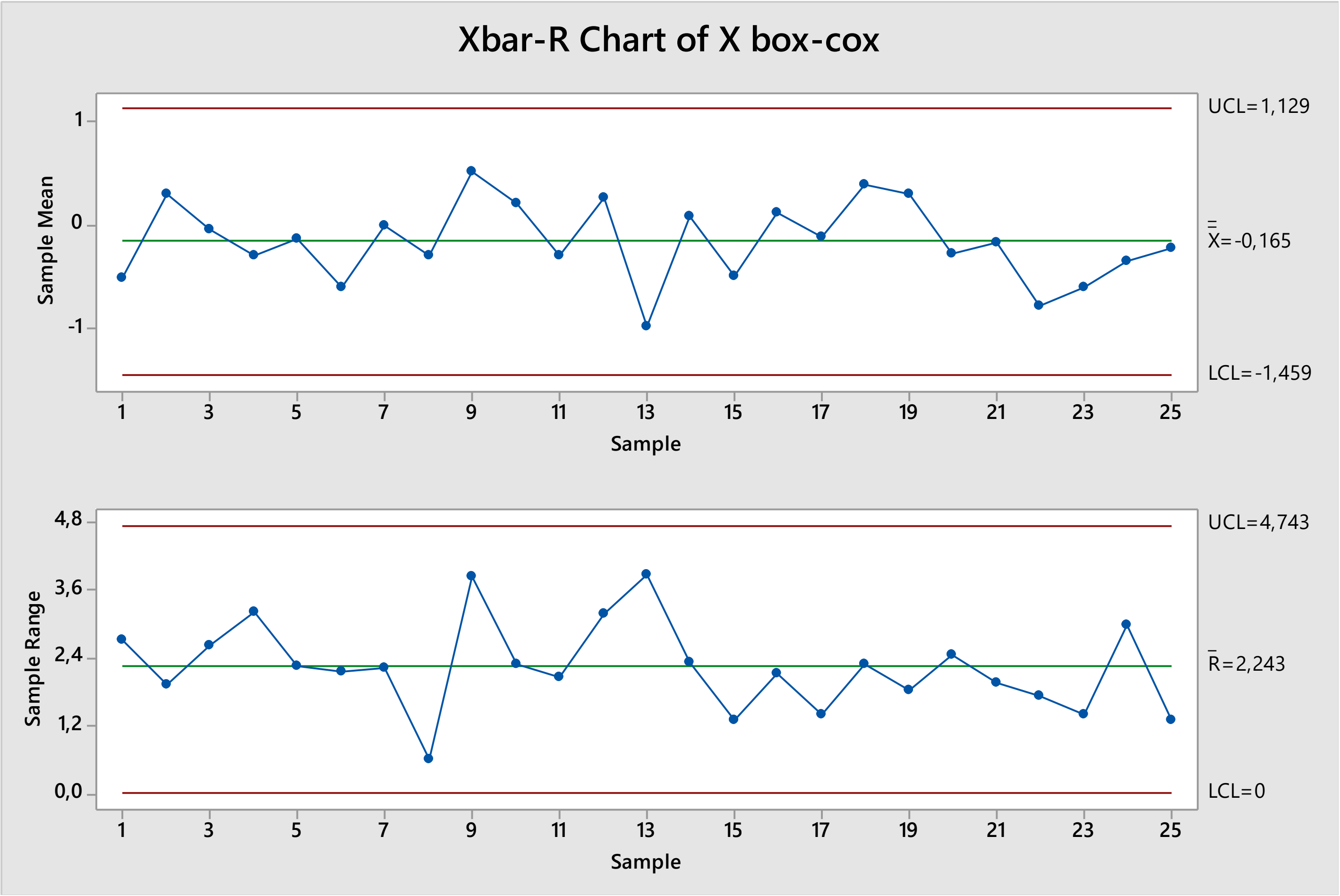
Normality is OK



P-value=0.191

EXERCISE 2 (SOLUTION)

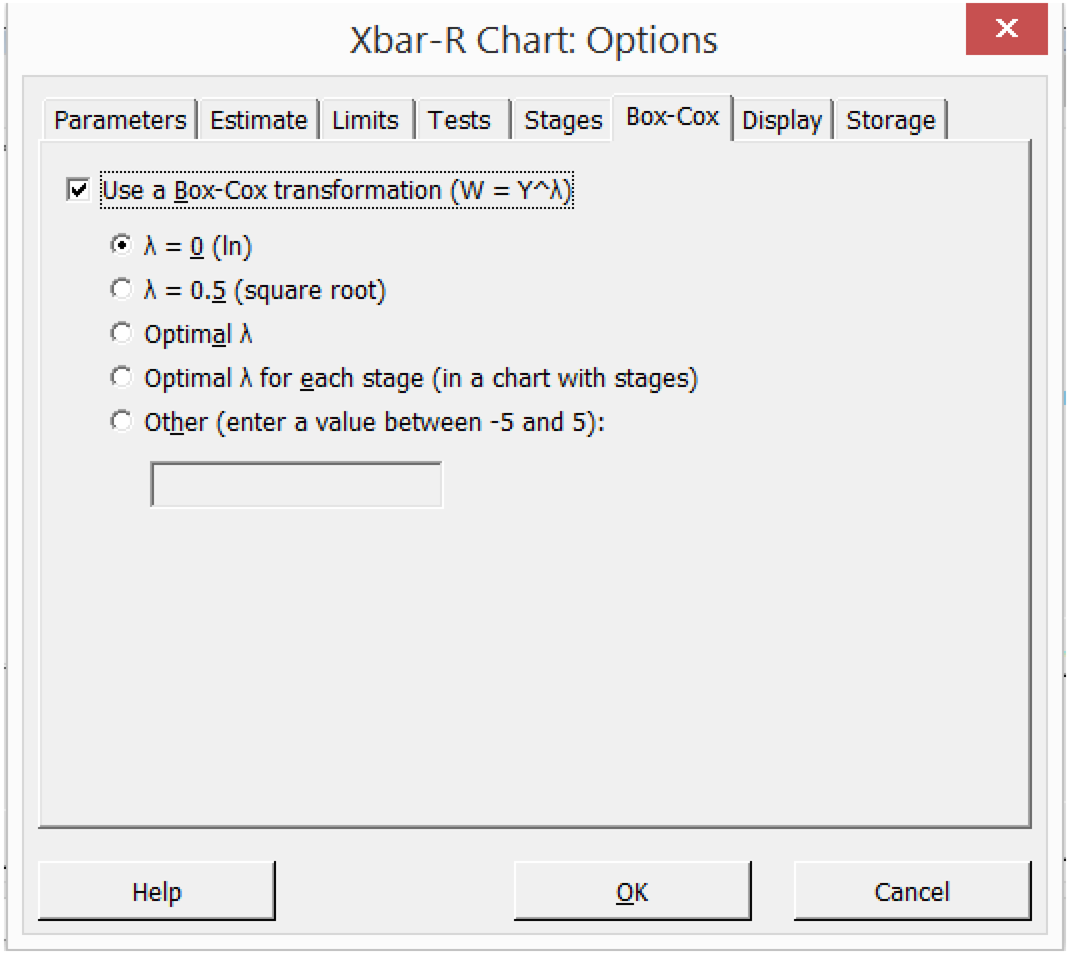
Control chart for transformed data:



The process is in control

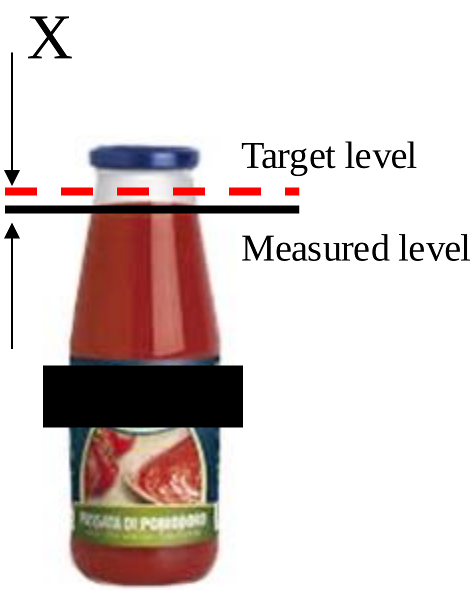
EXERCISE 2 (SOLUTION)

Minitab allows performing the Box-Cox transformation as an additional option for the control chart (one single step):



EXERCISE 3

A company that produces tomato sauce is testing a new accurate system to control the level of sauce in each bottle. To this aim, they are monitoring the level in each bottle by using an automated optical system that measures the deviation (in mm) of the sauce level from a target level. See *tomato_sauce.csv*.



The data for 300 consecutive measurements are reported in the provided table.

Since the measured data are auto-correlated, check if:

a) Data batching is suitable to get rid of the data auto-correlation

b) Data gapping is suitable to get rid of the data auto-correlation

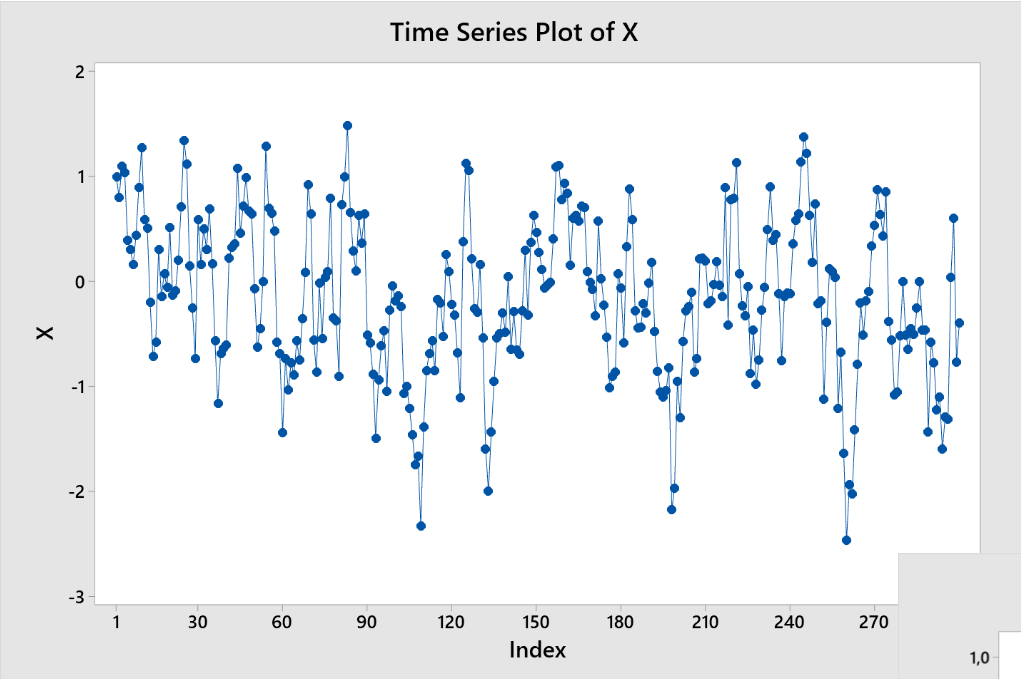
Comment the results

	C1
	X
1	1,00
2	0,80
3	1,10
4	1,04
5	0,40
6	0,30
7	0,16
8	0,44
9	0,90
10	1,27
11	0,59
12	0,51
13	-0,20
14	-0,72
15	-0,58
16	0,30
17	-0,14
18	0,07
19	-0,05
20	0,52
21	-0,13
22	-0,09
23	0,20
24	0,71
25	1,34

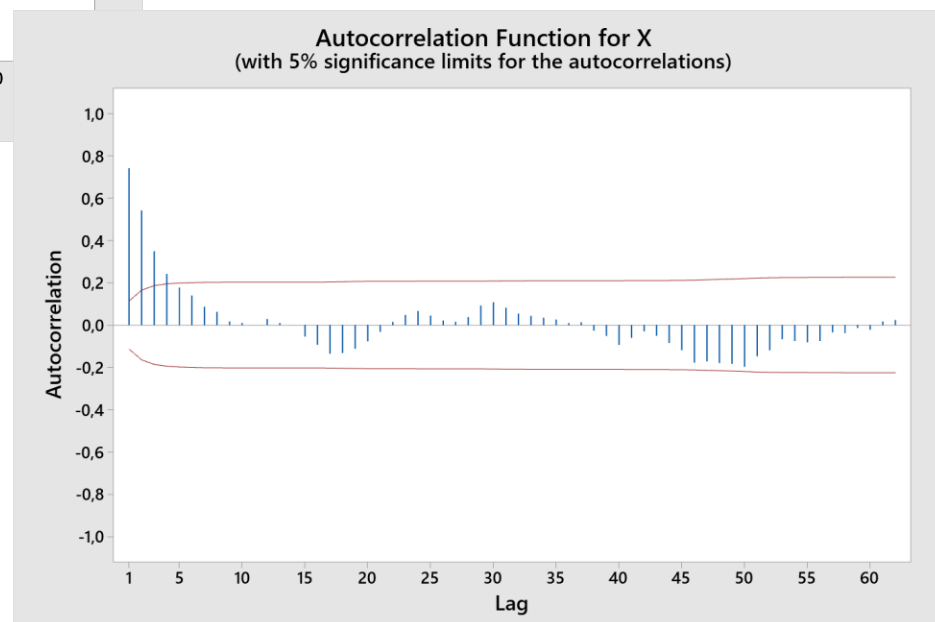
…

EXERCISE 3 (solution)

Time series plot



Auto-correlation function



Runs-test

Test

Null hypothesis H_0: The order of the data is random

Alternative hypothesis H_1: The order of the data is not random

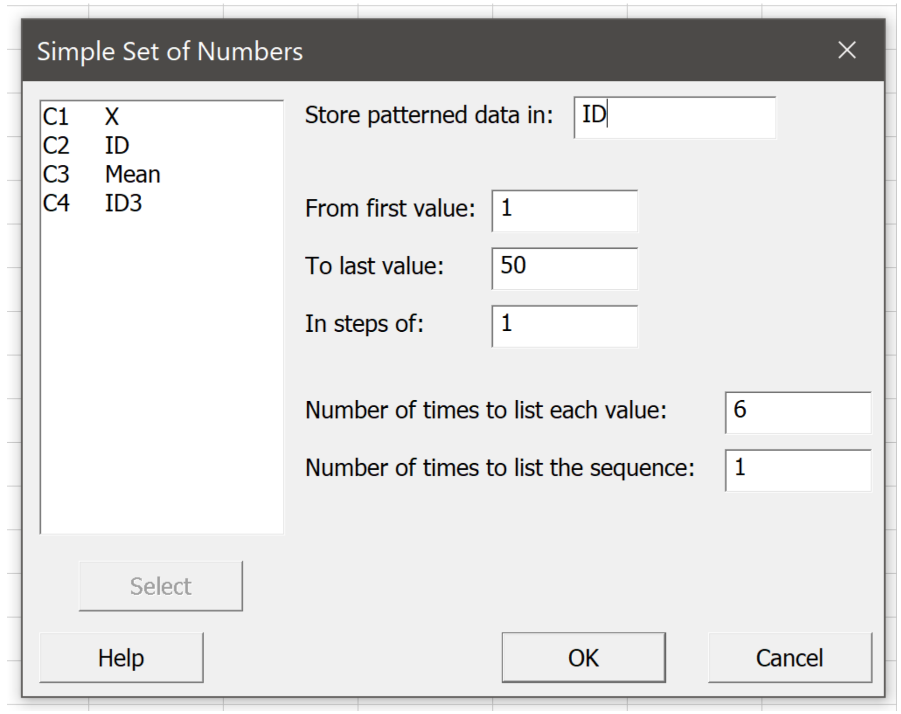
Number of Runs		
Observed	Expected	P-Value
66	151,00	0,000

EXERCISE 3 (solution)

a) Batching

	C1	C2	C3
	X	ID	Mean
7	0,16	2	-0,4250
8	0,44	2	0,7140
9	0,90	2	0,1330
10	1,27	2	-0,1450
11	0,59	2	-0,7915
12	0,51	2	-0,0200
13	-0,20	3	0,0050
14	-0,72	3	0,4330
15	-0,58	3	0,2530
16	0,30	3	-0,8310
17	-0,14	3	-0,3200
18	0,07	3	-1,3570
19	-0,05	4	-1,1120
20	0,52	4	-0,1280
21	-0,13	4	0,0770
22	-0,09	4	-0,3850
23	0,20	4	-0,9530
24	0,71	4	-0,4520
25	1,34	5	0,1960
26	1,12	5	0,1167
27	0,15	5	0,8190
28	-0,25	5	0,5544
29	-0,73	5	-0,0060
30	0,59	5	-0,5510
31	0,16	6	0,0840
32	0,50	6	-0,2090

Create an ID vector with «make patterned data»



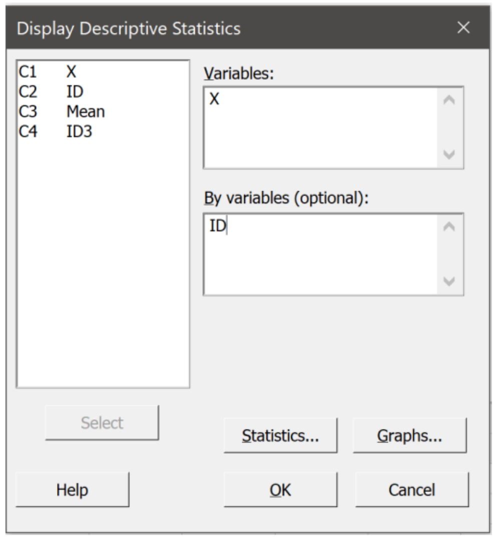
→ 50 subgroups of length 6

EXERCISE 3 (solution)

a) Batching

	C1	C2	C3
	X	ID	Mean
7	0,16	2	-0,4250
8	0,44	2	0,7140
9	0,90	2	0,1330
10	1,27	2	-0,1450
11	0,59	2	-0,7915
12	0,51	2	-0,0200
13	-0,20	3	0,0050
14	-0,72	3	0,4330
15	-0,58	3	0,2530
16	0,30	3	-0,8310
17	-0,14	3	-0,3200
18	0,07	3	-1,3570
19	-0,05	4	-1,1120
20	0,52	4	-0,1280
21	-0,13	4	0,0770
22	-0,09	4	-0,3850
23	0,20	4	-0,9530
24	0,71	4	-0,4520
25	1,34	5	0,1960
26	1,12	5	0,1167
27	0,15	5	0,8190
28	-0,25	5	0,5544
29	-0,73	5	-0,0060
30	0,59	5	-0,5510
31	0,16	6	0,0840
32	0,50	6	-0,2090

Create a vector of means for subgroups of length 6



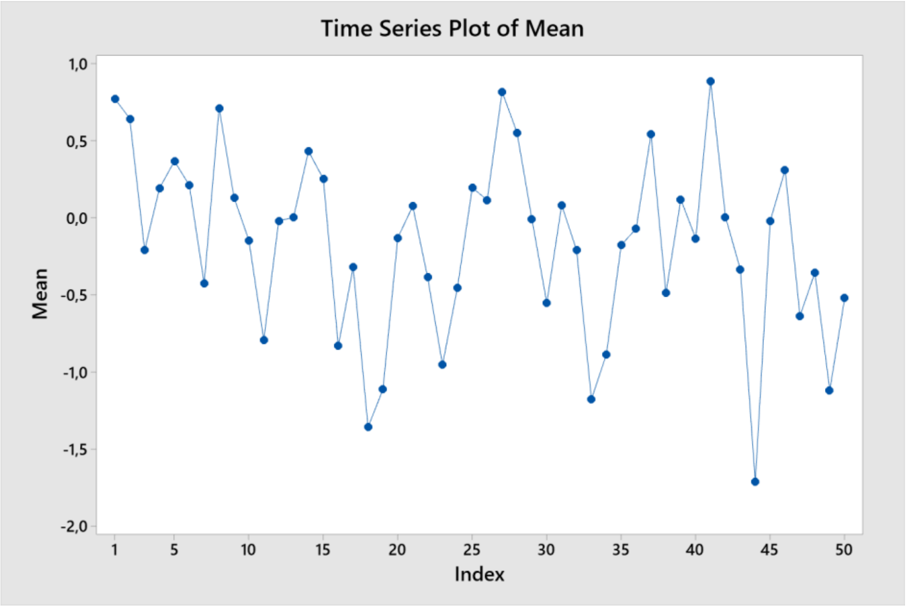
Statistics

Variable	ID	Mean
X	1	0,773
	2	0,645
	3	-0,209
	4	0,193
	5	0,369
	6	0,212
	7	-0,425
	8	0,714
	9	0,133
	10	-0,145
	11	-0,7915
	12	-0,020
	13	0,005
	14	0,433
	15	0,253

Store this column in the wroksheet (copy and paste)

EXERCISE 3 (solution)

a) Batching



Runs test:

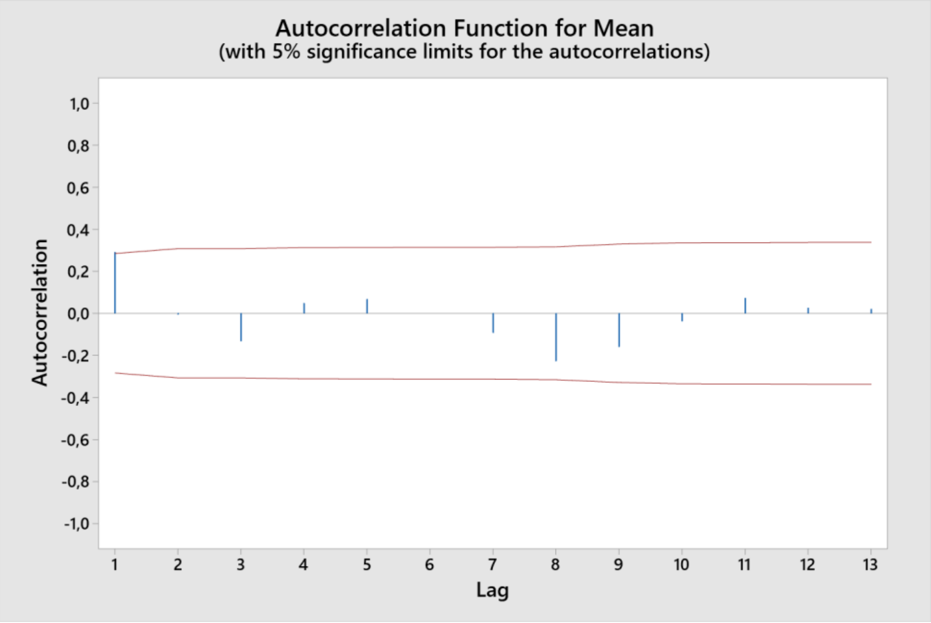
Test

Null hypothesis	H_0: The order of the data is random
Alternative hypothesis	H_1: The order of the data is not random

Number of Runs

Observed	Expected	P-Value
20	25,64	0,102

Accept randomness



Bartlett for lag 1:

$$\frac{z_{\alpha/2}}{\sqrt{n}} = \frac{1.96}{7.07} = 0.277$$

$$\boxed{|r_1| = 0.29 > \frac{z_{\alpha/2}}{\sqrt{n}}}$$

Reject!

EXERCISE 3 (solution)

b) Gapping

	C1	C2	C3	C4
	X	ID	Mean	IDgap
7	0,16	2	-0,4250	1
8	0,44	2	0,7140	0
9	0,90	2	0,1330	0
10	1,27	2	-0,1450	0
11	0,59	2	-0,7915	0
12	0,51	2	-0,0200	0
13	-0,20	3	0,0050	1
14	-0,72	3	0,4330	0
15	-0,58	3	0,2530	0
16	0,30	3	-0,8310	0
17	-0,14	3	-0,3200	0
18	0,07	3	-1,3570	0
19	-0,05	4	-1,1120	1
20	0,52	4	-0,1280	0
21	-0,13	4	0,0770	0
22	-0,09	4	-0,3850	0
23	0,20	4	-0,9530	0
24	0,71	4	-0,4520	0
25	1,34	5	0,1960	1
26	1,12	5	0,1167	0
27	0,15	5	0,8190	0
28	-0,25	5	0,5544	0
29	-0,73	5	-0,0060	0
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Create an ID vector that is:

- 0 for values excluded after gapping
- 1 for values kept in gapping

Easiest way is to create the vector in Excel and paste it in Minitab

EXERCISE 3 (solution)

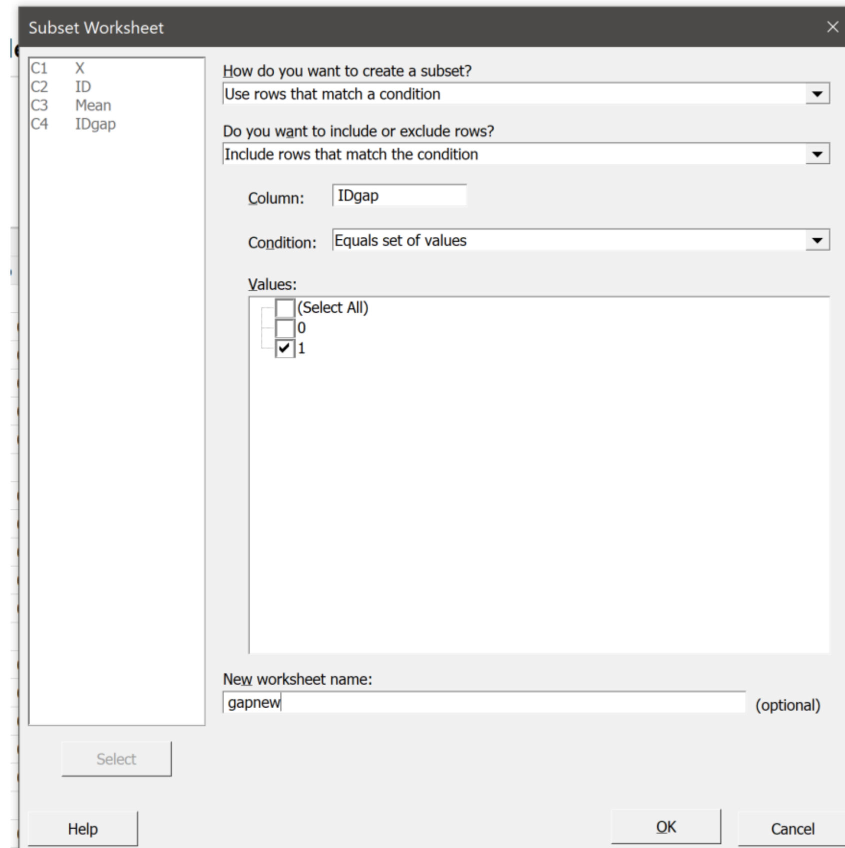
b) Gapping

	C1	C2	C3	C4
	X	ID	Mean	IDgap
7	0,16	2	-0,4250	1
8	0,44	2	0,7140	0
9	0,90	2	0,1330	0
10	1,27	2	-0,1450	0
11	0,59	2	-0,7915	0
12	0,51	2	-0,0200	0
13	-0,20	3	0,0050	1
14	-0,72	3	0,4330	0
15	-0,58	3	0,2530	0
16	0,30	3	-0,8310	0
17	-0,14	3	-0,3200	0
18	0,07	3	-1,3570	0
19	-0,05	4	-1,1120	1
20	0,52	4	-0,1280	0
21	-0,13	4	0,0770	0
22	-0,09	4	-0,3850	0
23	0,20	4	-0,9530	0
24	0,71	4	-0,4520	0
25	1,34	5	0,1960	1
26	1,12	5	0,1167	0
27	0,15	5	0,8190	0
28	-0,25	5	0,5544	0
29	-0,73	5	-0,0060	0

Create a new worksheet where you keep only the rows where IDgap = 1

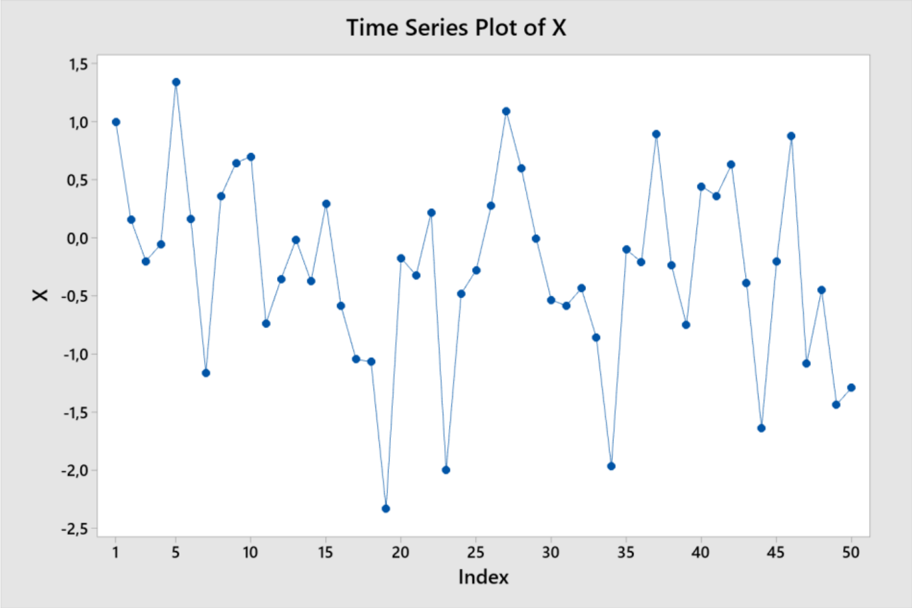
Data → subset worksheet

In the new worksheet, the column X has 50 rows (gapped version of original X)



EXERCISE 3 (solution)

b) Gapping



Runs test:

Test

Null hypothesis	H_0: The order of the data is random
Alternative hypothesis	H_1: The order of the data is not random

Number of Runs		
Observed	Expected	P-Value
20	25,96	0,088

Accept randomness (at 95%)

Bartlett for lag 1:

$$\frac{z_{\alpha/2}}{\sqrt{n}} = \frac{1.96}{7.07} = 0.277$$

$$\boxed{|r_1| = 0.232 < \frac{z_{\alpha/2}}{\sqrt{n}}}$$

Accept randomness at 95%